Q1 · 1 电力曳引式升降机 / Electric traction lift

驱动电动机带动曳引轮，通过与悬吊绳的摩擦，使轿厢和对重反向移动。下图采用典型的 1:1 无齿轮曳引布置。

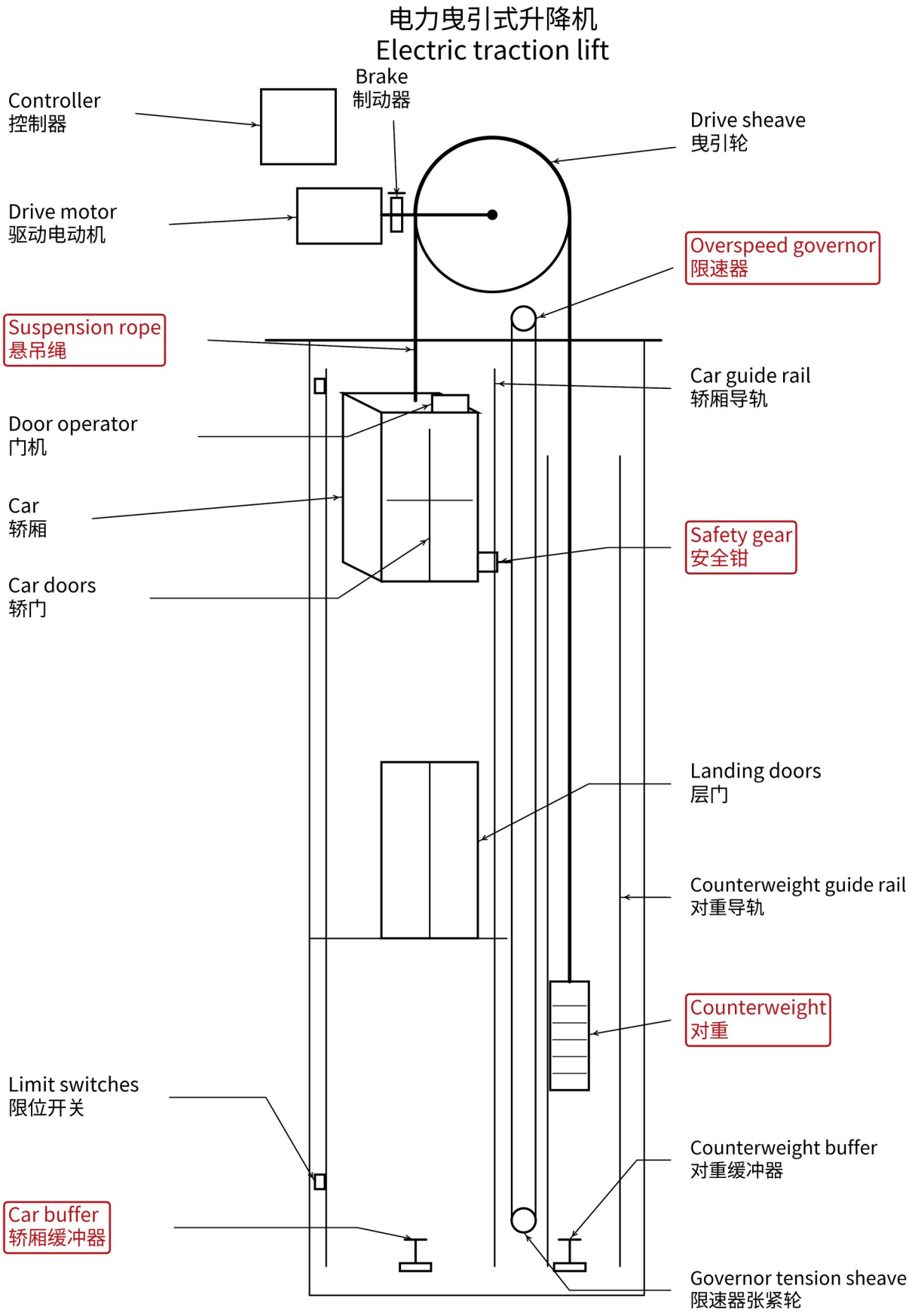
The Drive motor rotates the Drive sheave. Friction with the Suspension rope moves the Car and Counterweight in opposite directions. The schematic shows a typical 1:1 gearless traction arrangement.

主要部件 / Major components	运行功能 / Operational functions
驱动电动机 Drive motor	提供转动曳引轮的动力。 Supplies power to rotate the Drive sheave.
曳引轮 Drive sheave	通过与悬吊绳的摩擦传递驱动力。 Transfers driving force by friction with the Suspension rope.
悬吊绳 Suspension rope	悬吊轿厢和对重，使两者反向移动。 Suspends the Car and Counterweight, which move in opposite directions.
轿厢 Car	承载乘客或货物。 Carries passengers or goods.
对重 Counterweight	平衡轿厢及部分载荷，减小电动机需克服的不平衡负载。 Balances the Car and part of its load, reducing the unbalanced load on the Drive motor.
轿厢导轨；对重导轨 Car guide rail; Counterweight guide rail	分别引导轿厢和对重垂直运行，限制横向移动。 Guide the Car and Counterweight respectively and restrain lateral movement.
制动器 Brake	采用弹簧施闸、电力释放；使驱动系统停止并保持静止。 Is spring-applied and electrically released; stops and holds the drive system.
控制器 Controller	接收召梯指令，协调驱动、制动、停车、平层和开关门。 Receives calls and coordinates drive, braking, stopping, levelling and door operation.
门机；轿门；层门 Door operator; Car doors; Landing doors	门机驱动轿门，并在层站联动层门；门关闭轿厢及层站入口。层门联锁确保其关闭锁紧后才可正常运行。 The Door operator drives the Car doors and couples to the Landing doors at a landing. The doors guard both entrances; landing-door interlocks require closure and locking before normal travel.
限速器；限速器张紧轮 Overspeed governor; Governor tension sheave	限速器检测速度，超速时触发安全停止，必要时使安全钳动作；限速器张紧轮维持限速器绳路张紧。 The Overspeed governor detects overspeed, initiates stopping and applies the Safety gear if necessary. The Governor tension sheave keeps its rope loop tensioned.
安全钳 Safety gear	发生向下超速或悬吊失效时，夹紧轿厢导轨，使轿厢停止并保持静止。 Grips the Car guide rail to stop and hold the Car during downward overspeed or suspension failure.
限位开关 Limit switches	在行程端部触发停止保护，防止超越允许行程。 Initiate stopping protection at the ends of permitted travel.
轿厢缓冲器；对重缓冲器 Car buffer; Counterweight buffer	分别在轿厢或对重发生底部越程时吸收冲击能量。 Absorb impact energy during bottom overtravel of the Car or Counterweight respectively.

运行简述 / Brief operation

层门关闭锁紧并满足安全条件后，控制器协调制动器释放与驱动电动机运行。轿厢到站减速、平层并停止，制动器保持静止后开门；发生超速时，限速器触发安全停止，必要时使安全钳动作。

With the Landing doors closed and locked and safety conditions satisfied, the Controller coordinates Brake release and Drive motor operation. The Car slows, levels and stops; the Brake holds it before the doors open. Overspeed triggers the Overspeed governor and, when necessary, the Safety gear.



1:1 曳引布置；轿厢与对重反向移动。
1:1 traction arrangement; the Car and Counterweight move in opposite directions.
细箭头为标注线；限速器绳路与悬吊绳分开绘制。
Thin arrows are label leaders; the governor rope loop is separate from the Suspension rope.

2 直顶式液压升降机 / Direct-driven cylinder hydraulic lift

油泵将液压油经阀组送入埋地油缸，推动柱塞伸出并直接顶升轿厢。下图为埋地油缸布置，柱塞与轿厢之间没有悬吊绳传动。

The Pump supplies oil through the Valve block to the Cylinder in borehole, extending the Piston to raise the Car directly. The schematic shows an in-ground cylinder arrangement with direct support of the Car.

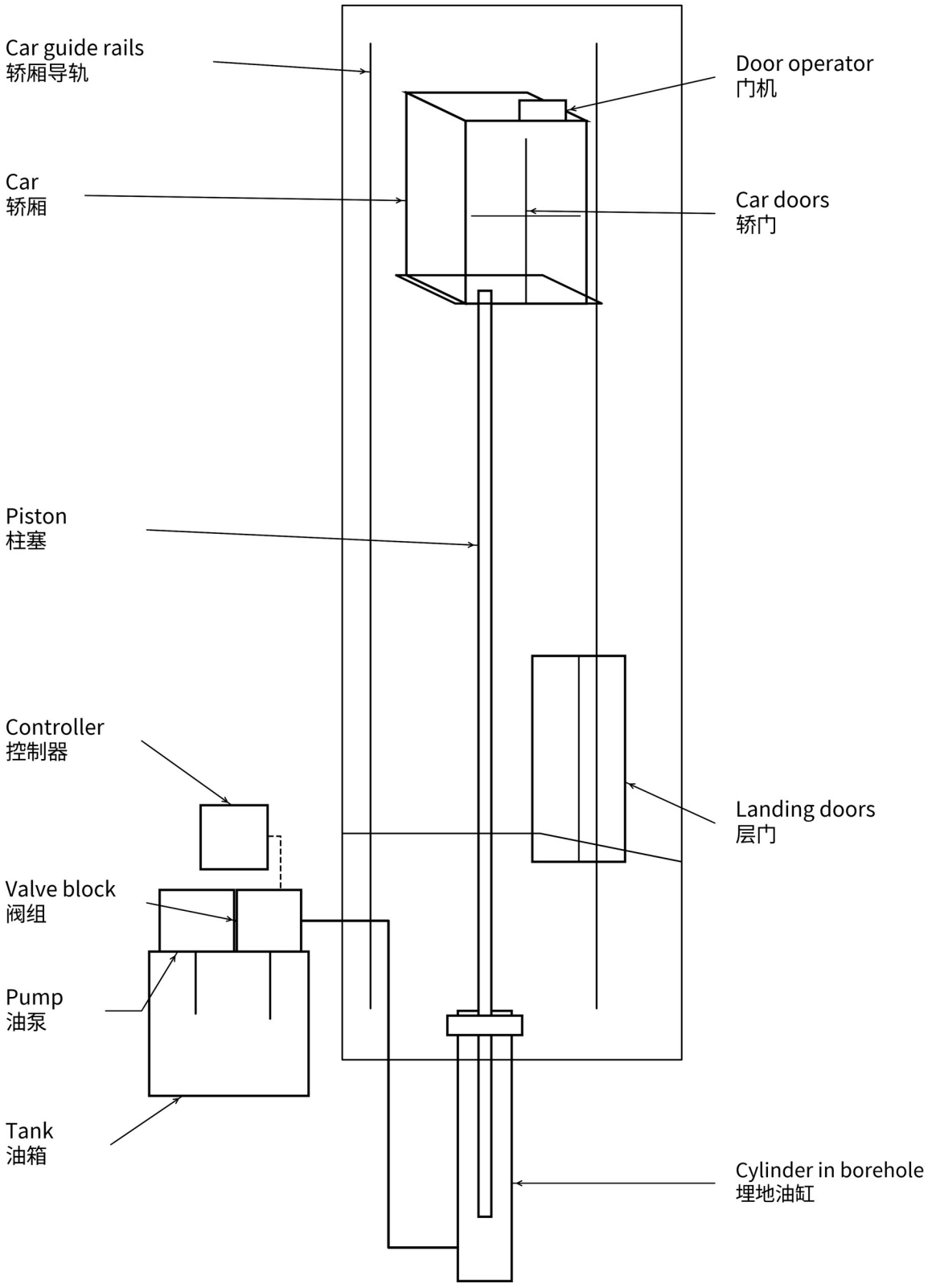
主要部件 / Major components	运行功能 / Operational functions
油箱 Tank	储存液压油，向油泵供油，并接收下降时的回油。 Stores hydraulic oil, supplies the Pump and receives returning oil during descent.
油泵 Pump	由电动机驱动，将油箱中的油送入埋地油缸，产生顶升所需油流。 Is electrically driven and delivers oil from the Tank to the Cylinder in borehole to raise the Car.
阀组 Valve block	控制进油、保压及回油，实现上升、停止和受控下降。 Controls oil admission, holding and return for ascent, stopping and controlled descent.
埋地油缸 Cylinder in borehole	容纳受压液压油；向地下延伸，为柱塞回缩提供所需空间。 Contains pressurised oil and extends below ground to accommodate Piston retraction.
柱塞 Piston	受油压作用伸出，直接顶升轿厢；回油时受轿厢重量作用而回缩。 Extends under oil pressure to raise the Car directly; retracts under Car weight as oil returns.
轿厢 Car	承载乘客或货物，由柱塞直接承托。 Carries passengers or goods and is directly supported by the Piston.
轿厢导轨 Car guide rails	引导轿厢垂直运行，限制横向移动。 Guide the Car vertically and restrain lateral movement.
控制器 Controller	接收召梯指令，协调油泵、阀组、停车、平层和开关门。 Receives calls and coordinates the Pump, Valve block, stopping, levelling and door operation.
门机；轿门；层门 Door operator; Car doors; Landing doors	门机驱动轿门，并在层站联动层门；门关闭轿厢及层站入口。层门联锁确保其关闭锁紧后才可正常运行。 The Door operator drives the Car doors and couples to the Landing doors at a landing. The doors guard both entrances; landing-door interlocks require closure and locking before normal travel.

运行简述 / Brief operation

上升：控制器启动油泵，油进入埋地油缸，使柱塞伸出。停止：油泵停止供油，阀组封住油液，使轿厢保持位置。下降：阀组允许油受控流回油箱，轿厢重量使柱塞回缩；平层后关闭回油通路，再开门。

Ascent: the Controller starts the Pump and oil extends the Piston. Holding: the Pump stops supplying oil and the Valve block retains oil to hold the Car. Descent: the Valve block meters oil back to the Tank and Car weight retracts the Piston. At the landing, return flow closes before the doors open.

直顶式液压升降机
Direct-driven cylinder hydraulic lift



Piston 直接承托 Car；油经 Valve block 进入或流出 Cylinder in borehole。
The Piston directly supports the Car; the Valve block controls oil flow into and out of the Cylinder in borehole.
结构和油路均为简化示意；细箭头仅为标注线。
Mechanical arrangement and oil path are simplified; thin arrows are label leaders.